

F619 — Thread 1 (timeboxed): the δ -quantization question resolves. Reading B (“the neutrino=0=identity, 0 has no phase $\rightarrow \delta$ absent”) is SEDUCTIVE but MIS-LOCATES δ . It is TRUE for the ν_1 Majorana phase (absent because $m_1=0$ — but that’s ALREADY banked, F589) and FALSE for δ_{PMNS} (the Dirac phase = the ν_2 - ν_3 relative phase, both nonzero $\rightarrow$ present & free). Reading A (δ quantized on the 127-cycle) is premise-contingent (fundamental discreteness, = the 127/128 premise) AND doesn’t select the value. δ_{PMNS} stays FREE (F618). No new result; the absent phase is ν_1 ’s Majorana phase (known), not δ . + Check 1 (ν S^4 at the rank-1 ridge): the two-barrier structure is solid; the precise locus placement needs closer analysis.

Lyra, Tue 2026-07-21 (Thread 1, timeboxed). Scrutinized Reading B hardest (it’s the prettiest, per the discipline). It resolves quickly and it’s a mis-location — the genuine content is a phase we already banked.

Check 2 — δ as a discrete log on $\text{GF}(128)^*$? (the load-bearing check)

The precise question that breaks it: WHICH phase? Majorana phase accounting for 3 neutrinos with $m_1=0$: - General 3×3 Majorana: 1 Dirac phase (δ) + 2 Majorana phases. - $m_1=0$ makes ONE Majorana phase unphysical (a massless Majorana state rephases freely). Physical phases: δ (Dirac) + 1 Majorana. (*Standard — this is exactly the F589 one-phase $0\nu\beta\beta$ sweep.*)

Reading B, correctly located: $\nu_1 = 0 =$ the additive identity = the massless state. Its Majorana phase is absent ($m_1=0$ removes it). So “the phase of the identity has none” is TRUE — for the ν_1 Majorana phase. But that is already banked (F589: $m_1=0$ $\Rightarrow$ one Majorana phase gone $\Rightarrow 0\nu\beta\beta$ is a one-phase sweep). Not a new result.

Reading B FAILS for δ_{PMNS} : the Dirac phase δ_{PMNS} is the relative phase of ν_2 and ν_3 — *both nonzero* ($m_2, m_3 \neq 0$). In discrete-log terms it’s $\sim \log(\nu_2) - \log(\nu_3)$ on $\text{GF}(128)^*$, and both ν_2, ν_3 have discrete logs (they’re nonzero elements). So δ_{PMNS} is well-defined (present), NOT the identity’s absent phase. Reading B conflates the massless- ν_1 Majorana phase (absent) with the Dirac δ (present) — that’s the seduction, and it doesn’t survive precise location.

Reading A (δ quantized on the 127-cycle): IF the substrate is fundamentally discrete (the code premise — *the same premise as 127/128, NOT derived*, F610/F613), then $\delta = \log(\nu_2) - \log(\nu_3) \in \{2\pi k/127\}$ — quantized ($\sim 2.83^\circ$ spacing, finer than DUNE $\rightarrow$ structural, not near-term testable). But (a) it rests on fundamental discreteness (premise-contingent), and (b) it quantizes δ but does not SELECT the value — δ is still free among 127 slots. So even Reading A keeps δ free (unselected) and is premise-contingent.

δ_{PMNS} stays FREE (F618). Neither reading changes it: Reading B applies to the (already-banked) ν_1 Majorana phase, not δ ; Reading A is premise-contingent and non-selecting.

Check 1 — does the chargeless $\Delta L=2$ S^4 neutrino locus sit at the rank-1 “ridge”?

- Solid (Casey’s structural anchor): rank-2 $\Rightarrow$ exactly two boundary barriers — the rank-1 orbit (“ridge”) + the Shilov boundary $S^4 \times S^1 / Z_2$ (“wall”). Casey drew the rank correctly. $\Rightarrow$
- Consistent: the massless ν_1 is the mode that escapes (tunnels/radiates out $\rightarrow$ free/continuous, F614) — matching “escapes past the ridge before the wall.” The escaping neutrino is ν_1 (massless, $=0$ /identity), which is *also* why its Majorana phase is absent (Check 2). So Checks 1 & 2 tie together: the escaping $\nu_1 =$ the identity = the phase-less massless mode.
- NEEDS CLOSE ANALYSIS (timebox limit): whether the chargeless S^4 *condensate* locus is precisely the rank-1 orbit (ridge) vs the Shilov S^4 (wall) — $Q=0$ on $S^4 \times S^1$ picks a sub-locus at fixed S^1 , and I can’t cleanly

place it (rank-1 orbit vs Shilov sub-locus) in the timebox. Banked as: needs close analysis; structural picture (two barriers, ν_1 escapes) holds; exact orbit-placement open.

Verdict (timeboxed, done)

- Reading B: mis-located. TRUE for ν_1 's Majorana phase (absent, $m_1=0$ — already banked); FALSE for δ_{PMNS} (Dirac, present, free). No new result. Caught, not sold.
- Reading A: premise-contingent + non-selecting. Quantizes δ only if the substrate is fundamentally discrete (=127/128 premise), and even then doesn't pick the value.
- δ_{PMNS} : FREE (F618 stands). The honest DUNE story is unchanged — BST predicts δ unconstrained.
- Check 1: two-barrier structure solid; ν_1 -escapes consistent; exact S^4 -locus placement needs close analysis.
- The genuine kernel Casey found: the massless neutrino ($=0=\text{identity}$) carries no phase — TRUE, and it IS a nice way to say " $m_1=0$ removes a Majorana phase." Worth stating in the paper *as the ν_1 Majorana phase*, not as δ . Moving to Thread 2 (quark hierarchy).

Tiers / handoffs

- δ_{PMNS} FREE (unchanged, F618). Reading B = the ν_1 Majorana phase (absent, $m_1=0$, banked F589) — real but not new, and NOT δ . Reading A = premise-contingent + non-selecting.
- @Cal — Reading B is TRUE only for the ν_1 Majorana phase ($m_1=0$, already banked), FALSE for δ_{PMNS} (mis-located Dirac phase); don't file " δ absent" — δ is present & free. Reading A rests on fundamental-discreteness (same premise as 127/128) and doesn't select the value. Clean "quantized $\neq$ selected $\neq$ free" distinction held.
- @Keeper — Thread 1 timeboxed-closed: δ stays FREE; Reading B mis-locates δ (it's the ν_1 Majorana phase, banked); Reading A premise-contingent. Check 1: two-barrier structure solid, exact S^4 -locus placement = needs-close-analysis (bank, don't block). Frame the "identity has no phase" correctly (ν_1 Majorana, not δ) if it goes in the paper.
- @Elie — nothing to overturn; δ free confirmed in phase-accounting terms. I'm joining Thread 2 (up-type tower) now.
- @Grace — δ_{PMNS} render: FREE (unchanged); the "massless- $\nu=\text{identity}=\text{no-phase}$ " note is the ν_1 Majorana phase (F589), not δ_{PMNS} — don't conflate on the render.

Notes only; no toys/theorems claimed. — Lyra